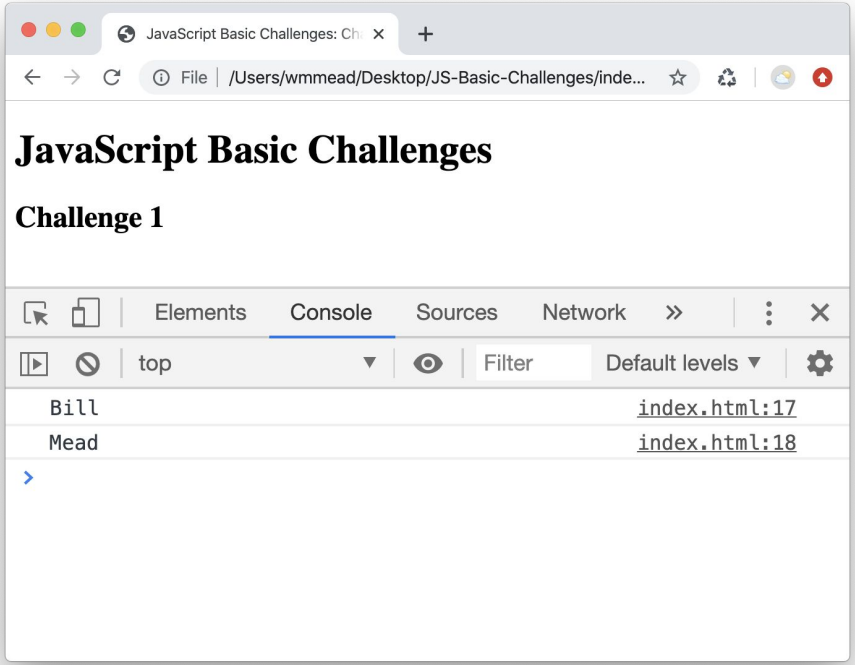
JavaScript Basic Challenges

01-09

Challenge 1

Create an HTML file and set the title to "JavaScript Basic Challenges: Challenge 1" Then add an <h1> and <h2> with that information.

Then add a script tag before the closing </body> tag and between the script tags create two variables. Assign your first name to one and your last name to the other, then make JavaScript display the value of each variable in the console.



Challenge 1 Answer

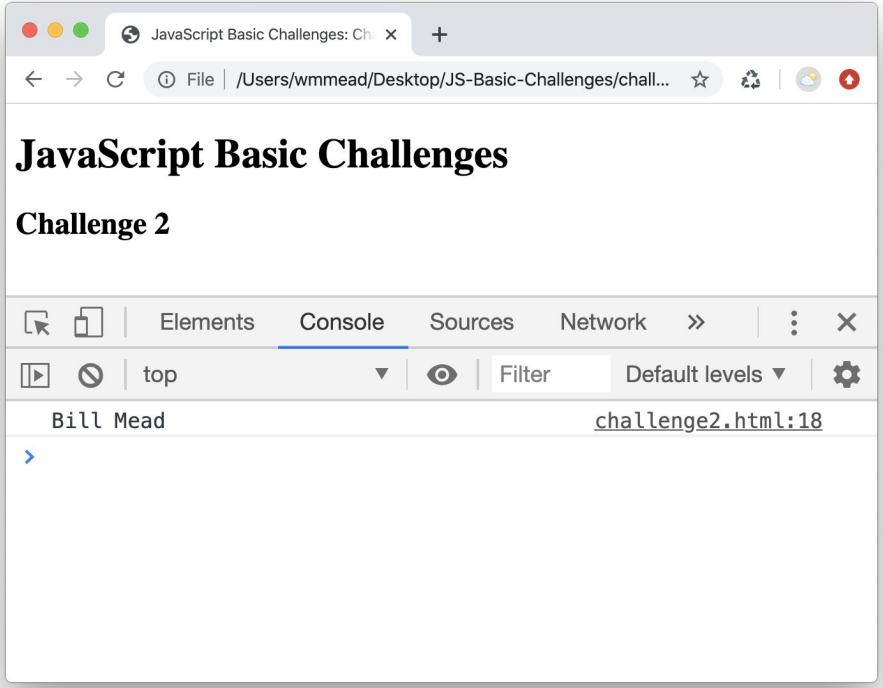
Your file should look pretty much like this:

```html
<!doctype html>
<html>
<head>
<meta charset="UTF-8">
<title>JavaScript Basic Challenges: Challenge 1</title>
</head>

<body>

    <h1>JavaScript Basic Challenges</h1>
    <h2>Challenge 1</h2>

    <script>

        var firstName = "Bill";
        var lastName = "Mead";
        console.log(firstName);
        console.log(lastName);

    </script>

</body>
</html>
```

Challenge 2

Change the title and the headings on the challenge1.html file and save it as challenge2.html.

Use plus signs (+) and your two variables to write your first name and last name to the console on one line with a space between your names.



Challenge 2 Answer

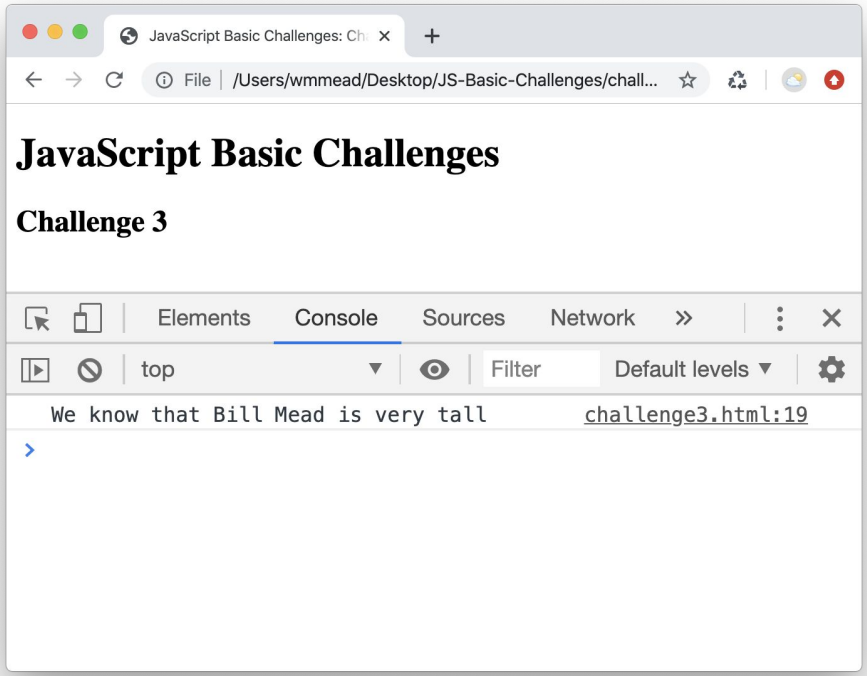
Your file should look pretty much like this:

```
<!doctype html>
<html>
<head>
<meta charset="UTF-8">
<title>JavaScript Basic Challenges: Challenge 2</title>
</head>

<body>

    <h1>JavaScript Basic Challenges</h1>
    <h2>Challenge 2</h2>

    <script>

        var firstName = "Bill";
        var lastName = "Mead";
        console.log(firstName + " " + lastName);

    </script>

</body>
</html>
```

Challenge 3

Update and save the file again, this time as challenge3.html

Add one more variable called "adjective" and assign it a value that describes you. Use the console to display a sentence that includes all three variables. This time use a template literal (tick marks, and ${} around variables).



Challenge 3 Answer

Your file should look pretty much like this:

```
<!doctype html>
<html>
<head>
<meta charset="UTF-8">
<title>JavaScript Basic Challenges: Challenge 3</title>
</head>

<body>

    <h1>JavaScript Basic Challenges</h1>
    <h2>Challenge 3</h2>

    <script>

        var firstName = "Bill";
        var lastName = "Mead";
        var adjective = "tall"
        console.log(`We know that ${firstName}
        ${lastName} is very ${adjective}`);

    </script>

</body>
</html>
```

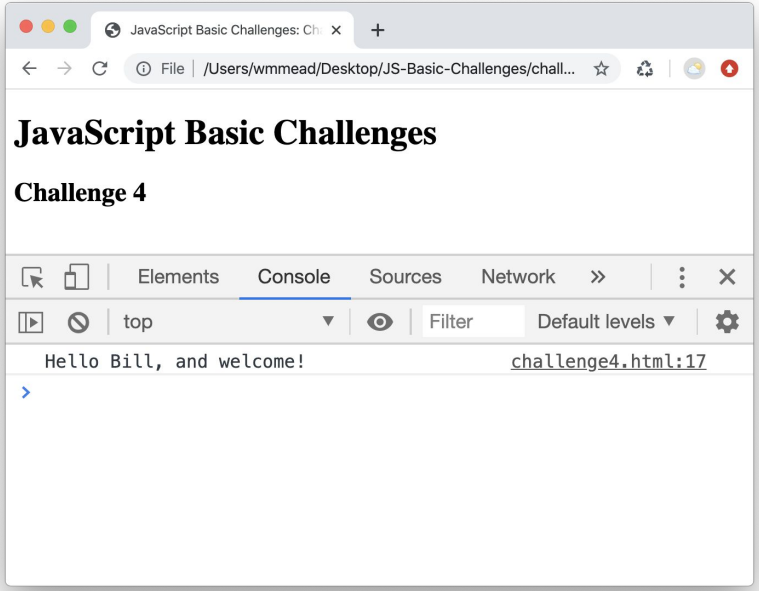
Challenge 4

Update and save the file again, this time as challenge4.html

In JavaScript, **prompt()** can be used like this:

var name = prompt("what's your name?", "Bob");
When the page loads a prompt will appear and the user can add some text. "Bob" is the default text. If you don't want default text, you can leave the second set of quotes empty.

Write script that prompts the user for their name, and then display a sentence in the console that greets the user by name and welcomes them.



Challenge 4 Answer

Your file should look pretty much like this.

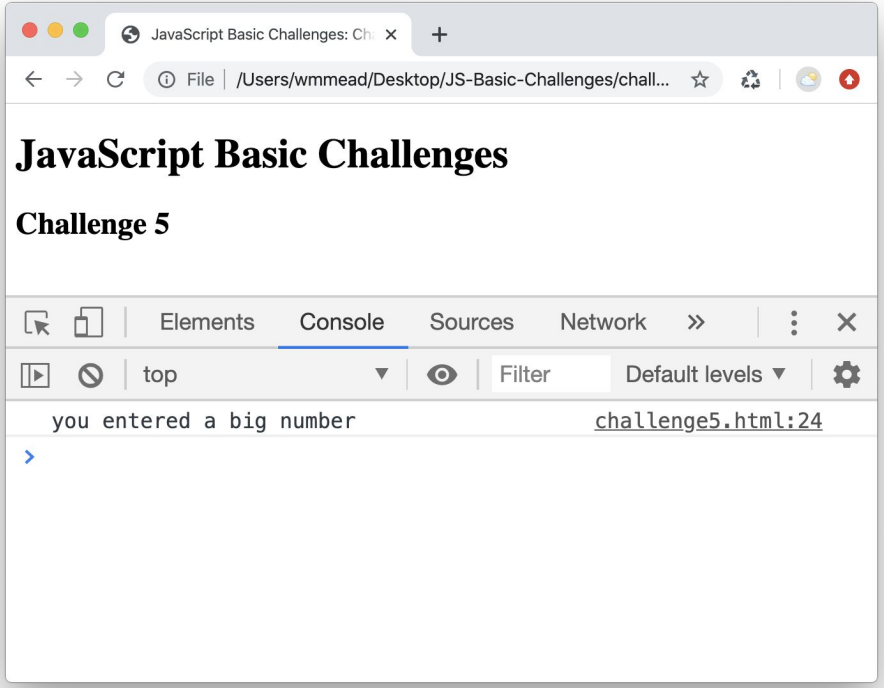
Notice the use of the tick marks.

```
<h1>JavaScript Basic Challenges</h1>
<h2>Challenge 4</h2>

<script>

    var name = prompt("What's your name?", "");
    console.log(`Hello ${name}, and welcome!`);

</script>
```

Challenge 5

Update and save the file again, this time as challenge5.html

This time, use prompt and ask the user for a whole number. If they type in a number less than 10, have the console log report, "You entered a number smaller than 10", otherwise, send "You entered a big number" to the console.



Challenge 5 Answer

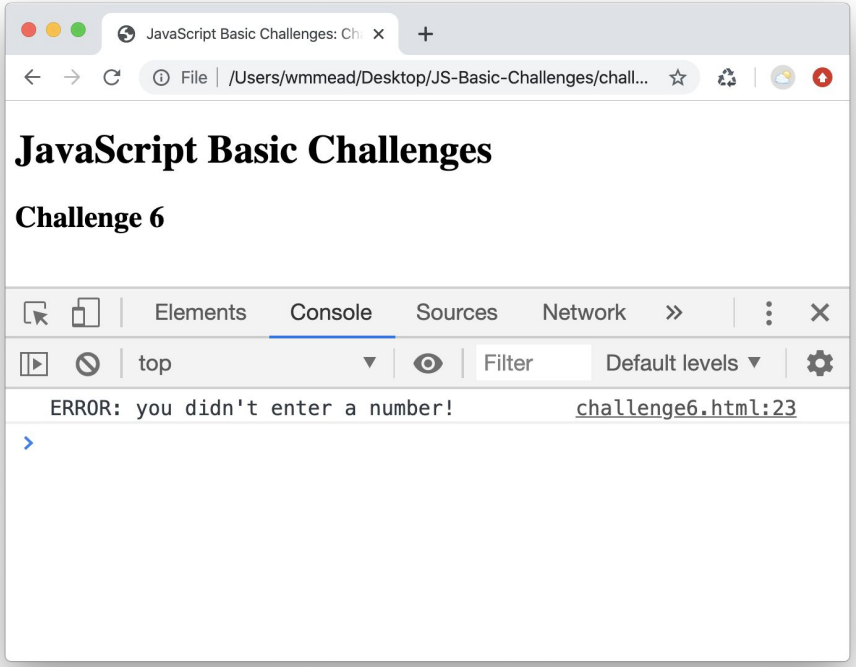
Your file should look pretty much like this:

```
var userNum = prompt("Enter a whole number", "");

if(userNum < 10){

    console.log("you entered a number less than 10");

} else {

    console.log("you entered a big number");

}
```

Challenge 6

Update and save the file again, this time as challenge6.html

Notice in challenge 5, if a user types something other than a number, the console tells us they entered a big number. In this challenge, fix that so that if a user types something other than a number, the console displays a message that says there was an error.



Challenge 6 Answer

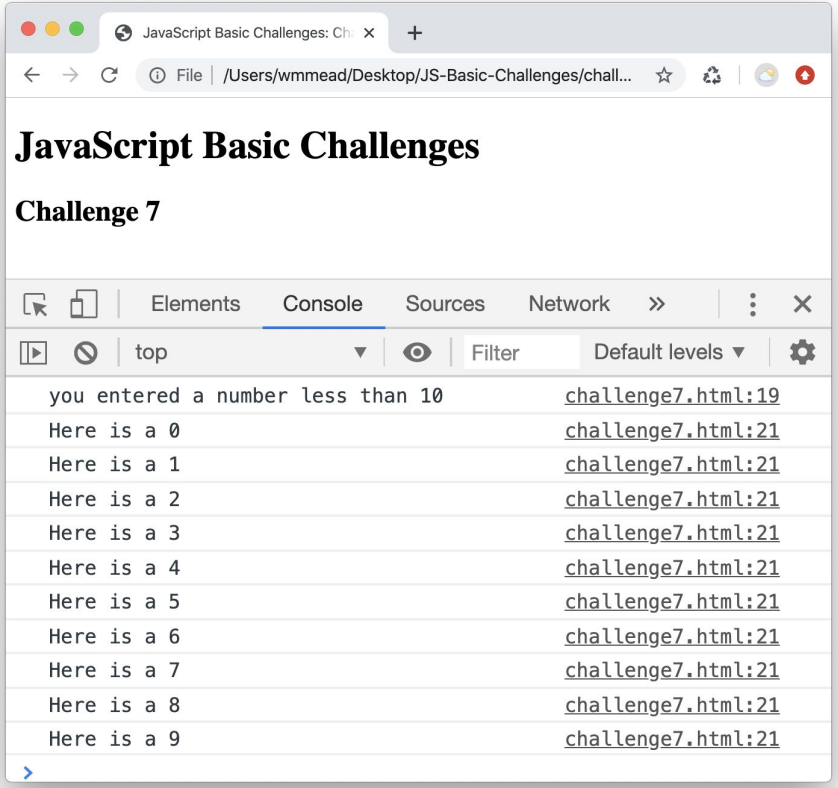
Your file should look pretty much like this:

```
var userNum = prompt("Enter a whole number", "");

if(userNum < 10){
    console.log("you entered a number less than 10");
} else if( userNum > 9) {
    console.log("you entered a big number");
} else {
    console.log("ERROR: you didn't enter a number!");
}
```

Challenge 7

Update and save the file again, this time as challenge7.html

For this challenge, if the user types a number less than 10, use a loop to print out the numbers from zero to 9 in the console.



Challenge 7 Answer

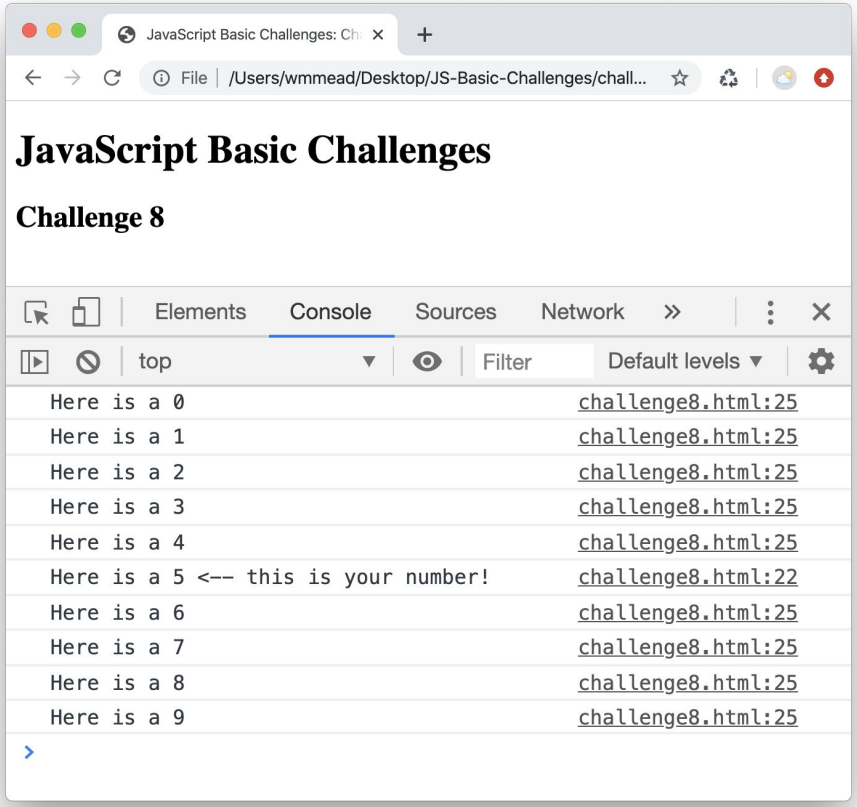
Your file should look pretty much like this:

```
var userNum = prompt("Enter a whole number", "");

if(userNum < 10){
    console.log("you entered a number less than 10");
    for( var i=0; i<10; i++){
        console.log(`Here is a ${i}`);
    }

} else if( userNum > 9) {
    console.log("you entered a big number");
} else {
    console.log("ERROR: you didn't enter a number!");
}
```

Challenge 8

Update and save the file again, this time as challenge8.html

For this challenge, if the user types a number less than 10, update the loop from the last challenge so that the console indicates the number the user entered.



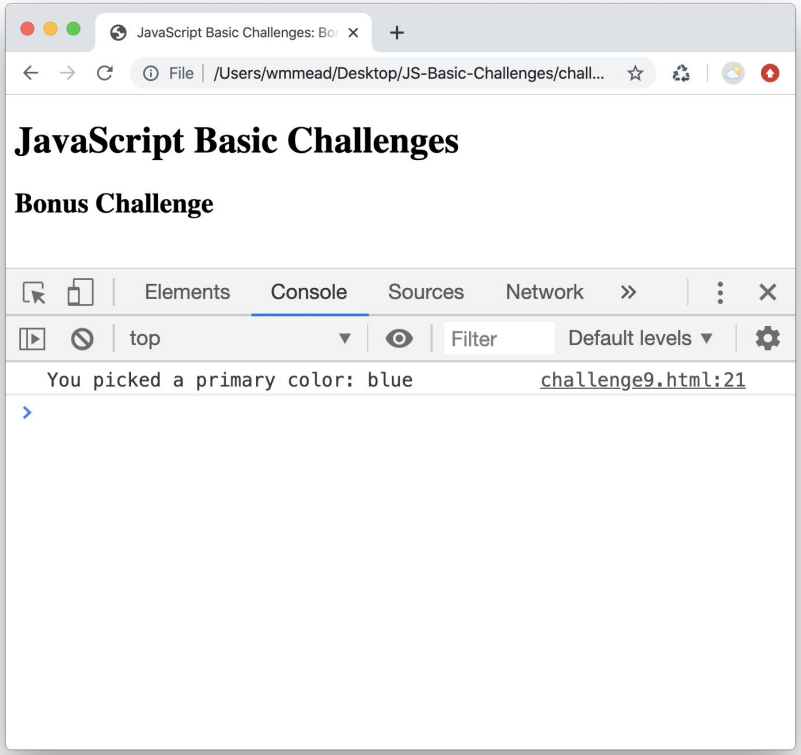
Challenge 8 Answer

Your file should look pretty much like this:

```
var userNum = prompt("Enter a whole number", "");

if(userNum < 10){
    console.log("you entered a number less than 10");
    for( var i=0; i<10; i++){
        if( i == userNum){
            console.log(`Here is a ${i} <-- this is your number!`);
        }
        else {
            console.log(`Here is a ${i}`);
        }
    }

} else if( userNum > 9) {
    console.log("you entered a big number");
} else {
    console.log("ERROR: you didn't enter a number!");
}
```

Bonus Challenge

Prompt the user for a color, then have the console report if the user typed a primary color (red, blue, or yellow) or a secondary color (orange, green or purple) or none of these.

Handle the situation for both uppercase and lowercase letters in the color the user types in.

Hint: remember the || means 'or'